9.9	**Osmoregulation and temperature regulation**
i	Know the gross and microscopic structure of the mammalian kidney.
ii	Understand how urea is produced in the liver from excess amino acids (details of the ornithine cycle are not required) and how it is removed from the bloodstream by ultrafiltration.
iii	Understand how solutes are selectively reabsorbed in the proximal tubule and how the Loop of Henle acts as a counter-current multiplier to increase the reabsorption of water.
iv	Understand how the pituitary gland and osmoreceptors in the hypothalamus, combined with the action of antidiuretic hormone (ADH) bring about negative feedback control of mammalian plasma concentration.
v	Understand how the kidney of a kangaroo rat (*Dipodomys sp*.) is adapted for life in a dry environment.
vi	Understand that an endotherm is able to produce heat through metabolic processes but an ectotherm must rely on the external environment.
vii	Understand how an endotherm is able to regulate its temperature through behaviour, and also physiologically through the autonomic nervous system, including the role of thermoreceptors, hypothalamus and the skin.